

Diabetes Prediction Using Machine Learning Classification Models

Summer Internship Project Report

Submitted By

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1 Abstract

Diabetes is one of the most common chronic diseases affecting millions of individuals worldwide. Early detection plays a significant role in reducing complications and improving patient outcomes. The objective of this project is to develop and compare multiple machine learning classification models capable of predicting diabetes based on patient medical information.

The Pima Indian Diabetes Dataset was used for this study. Three machine learning algorithms, namely Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM), were trained and evaluated. Data preprocessing techniques such as train-test splitting and feature scaling were performed before model development. The models were compared using Accuracy, Precision, Recall, and F1-Score. Experimental results showed that the SVM model achieved the highest overall performance among the selected classifiers. The study demonstrates the effectiveness of machine learning techniques in supporting early diagnosis of diabetes.

2 Introduction

Diabetes mellitus is a metabolic disorder characterized by elevated blood glucose levels caused by insufficient insulin production or ineffective insulin utilization. According to global healthcare reports, diabetes continues to be a major public health challenge and contributes significantly to mortality and healthcare costs.

Machine Learning has emerged as an effective tool for healthcare prediction tasks. By analyzing historical patient data, machine learning algorithms can identify hidden patterns and assist healthcare professionals in disease diagnosis. Predictive systems can reduce diagnosis time, improve decision-making, and support preventive healthcare strategies.

The objective of this project is to compare different classification algorithms and determine which model performs best for diabetes prediction.

3 Dataset Description

The Pima Indian Diabetes Dataset contains diagnostic information collected from female patients.

The dataset consists of 768 records and 8 input features:

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness

- Insulin
- BMI
- Diabetes Pedigree Function
- Age

Target Variable:

- Outcome = 0 (Non-Diabetic)
- Outcome = 1 (Diabetic)

4 Methodology

The project follows a standard machine learning workflow consisting of Data Loading, Data Inspection, Feature Selection, Train-Test Split, Feature Scaling, Model Training, and Model Evaluation. The dataset was divided into training and testing sets using an 80:20 ratio. StandardScaler was applied to normalize feature values before training the classification models.

5 Classification Algorithms

5.1 Logistic Regression

$$P(Y = 1|X) = \frac{1}{1 + e^{-z}} \quad (1)$$

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \quad (2)$$

5.2 K-Nearest Neighbors (KNN)

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (3)$$

5.3 Support Vector Machine (SVM)

$$f(x) = w^T x + b \quad (4)$$

$$y_i(w^T x_i + b) \geq 1 \quad (5)$$

6 Evaluation Metrics

6.1 Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

6.2 Precision

$$Precision = \frac{TP}{TP + FP} \quad (7)$$

6.3 Recall

$$Recall = \frac{TP}{TP + FN} \quad (8)$$

6.4 F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (9)$$

7 Experimental Results

Table 1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.714286	0.608696	0.518519	0.560000
KNN	0.701299	0.583333	0.518519	0.549020
SVM	0.720779	0.622222	0.518519	0.565657

The results indicate that the SVM classifier achieved the highest overall performance among the evaluated models.

8 Confusion Matrix Analysis

8.1 Logistic Regression

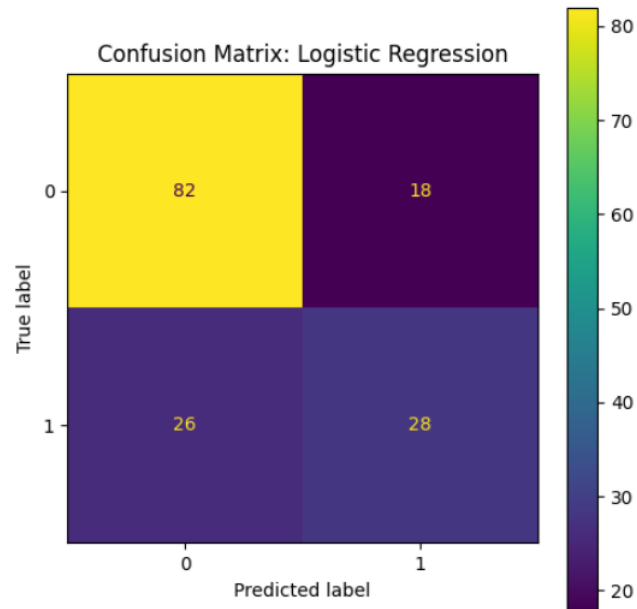


Figure 1: Confusion Matrix for Logistic Regression

8.2 K-Nearest Neighbors

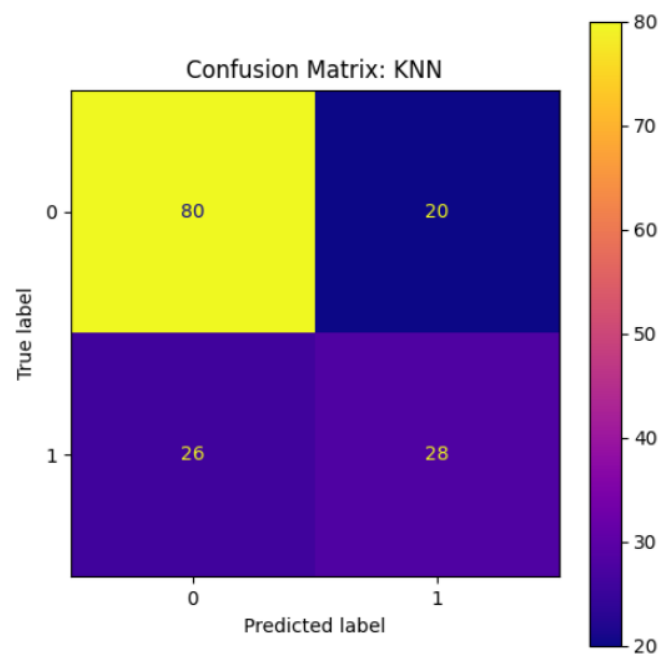


Figure 2: Confusion Matrix for KNN

8.3 Support Vector Machine

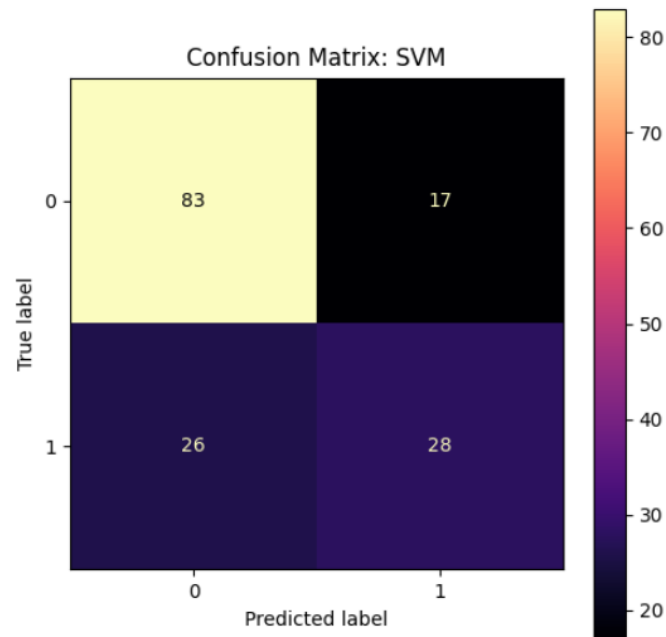


Figure 3: Confusion Matrix for SVM

9 Discussion

The comparative analysis demonstrates that all three machine learning algorithms are capable of predicting diabetes using patient medical measurements. Logistic Regression provided interpretable predictions and served as a strong baseline model. KNN achieved competitive performance but was more sensitive to feature scaling and neighborhood selection. SVM produced the highest overall performance and demonstrated superior classification capability.

Feature scaling played an important role in improving model performance. Since KNN and SVM are distance-based algorithms, standardization ensured that all features contributed equally to the learning process. The confusion matrices further highlighted the strengths and limitations of each classifier.

10 Conclusion

This project successfully implemented and evaluated Logistic Regression, KNN, and SVM for diabetes prediction using the Pima Indian Diabetes Dataset. Based on the obtained results, SVM achieved the best overall performance with an accuracy of approximately 72.08% and an F1-score of 56.57%.

The findings demonstrate that machine learning techniques can effectively assist healthcare professionals in disease prediction and early diagnosis. Future work can include ensemble

learning methods, deep learning approaches, hyperparameter optimization, and deployment as a healthcare decision-support application.

11 References

1. Pima Indians Diabetes Dataset.